



Iceland glacier melts rapidly

Comparison of pictures of a glacier in Iceland in the 80s and present-day drone pictures. Check all of them here. <https://dgit.in/nov19-41>



Racist algorithm

A widely-used healthcare algorithm in the US reportedly responds to the healthcare needs of white people first. <https://dgit.in/nov19-42>

standing was signed between KAGRA, LIGO and Virgo to conduct searches for gravitational waves together, and share data. The agreement also allows for future collaborators to join in the global network of gravitational wave observatories. The MOU supersedes existing agreements of the LIGO-Virgo collaboration. David Reitze, executive director of the LIGO Laboratory said, “This is a great example of international scientific cooperation. Having KAGRA join our network of gravitational-wave observatories will significantly enhance the science in the coming decade.”

KAGRA is expected to increase the detection rate of collision events that lead to the creation of gravitational ripples across spacetime. These include interactions between massive and exotic bodies such as neutron stars and black holes, among the most energetic phenomena in the universe. Initially, the KAGRA detector will operate at sensitivities that are too low to detect gravitational waves, but this performance will be gradually improved based on instrumentation. By the end of December 2019, KAGRA is expected to join in on the third observation run of the LIGO-Virgo collaboration, which began on April 1, 2019. All the detectors will cooperate to simultaneously participate in observation runs. Considering that the systems are improved between the runs, the observatories cannot search for gravitational waves round the clock.

As a bonus, the German-British Collaboration for the Detection of Gravitational Waves (GEO), which has



A cross section illustration of the underground KAGRA detector

a 600 metre long detector in Germany called the GEO600, also joined the collaboration on 4 October. The instrument is not powerful enough to directly participate in the detection of gravitational waves from distant collisions, but was instrumental in validating key technologies for future gravitational wave detectors.

All the three collaborations are truly massive. The LIGO project also includes the GEO collaboration, has over 1,300 scientists from more than 90 institutions in over 15 countries around the world. The Virgo collaboration is made up of over 480 scientists from 8 European countries. The KAGRA collaboration is composed of over 360 individuals from 100 institutions in almost 15 countries. Gravitational wave astronomy is truly a colossal effort on a global scale.

KAGRA, along with LIGO, Virgo and GEO600 are the second generation of gravitational

wave detectors, forming an international network through cooperation. There is only one second generation observatory to be added to the network, and that one is located in India. The IndiGO Indian Initiative in Gravitational-wave Observations (IndiGO) consortium has partnered with the LIGO laboratory

for the project. A site near Hingoli in Maharashtra has been selected for the detector, and the observatory is expected to be commissioned by 2024. The IndiGO observatory is expected to improve source localisation, which means researchers can more accurately pinpoint the origin of gravitational waves in the sky. This is possible if all the detectors observe the same event simultaneously, through triangulation. The first detection of gravitational waves by a black hole merger by LIGO in 2015, for example, could only be narrowed down to a broad portion of the southern hemisphere sky. The 2017 detection by the LIGO-Virgo collaboration was able to pinpoint the location to 0.1 percent of the sky, allowing other astronomical instruments to jump at the opportunity and observe the galaxy in visible light as well. The discovery opened up a new era of multi-messenger astronomy, where many observatories simultaneously track the same event. When KAGRA joined the collaboration, Reitze said, “the more detectors we have in the global gravitational-wave network, the more accurately we can localise the gravitational-wave signals on the sky, and the better we can determine the underlying nature of cataclysmic events that produced the signals”. The IndiGo observatory is expected to improve localisation by more than an order of magnitude. **4**



The cryostat housing the cryogenic mirror for making the gravitational wave observations at KAGRA